

Blood Sugar Battle

Teacher Key + Manual Scoring Guide

Format: 12 questions, 31 possible points. Groups write first, then score after you reveal the answer.

Round: Hormone Wars

Question	1 (2 pts) - Sugary cereal raises blood glucose quickly. Which hormone increases, and what is its overall effect?
Correct answer	Insulin increases; it promotes glucose uptake/storage and lowers blood glucose.
Scoring	1 pt insulin; 1 pt lowers blood glucose through uptake/storage.

Question	2 (2 pts) - Several hours after lunch, blood glucose begins to fall. Which hormone increases, and what does it do?
Correct answer	Glucagon increases; it raises blood glucose, primarily by stimulating breakdown of liver glycogen.
Scoring	1 pt glucagon; 1 pt raises blood glucose / stimulates liver glycogen breakdown.

Question	3 (2 pts) - Which pancreatic cell type releases insulin: alpha cells or beta cells?
Correct answer	Beta cells.
Scoring	2 pts beta cells.

Round: Where Does It Go?

Question	4 (3 pts) - After a carbohydrate-containing meal, name three major places or pathways glucose can go next.
Correct answer	Immediate fuel; glycogen storage in liver/muscle; fat storage when excess remains.
Scoring	1 pt each: immediate fuel, glycogen storage, fat storage.

Question	5 (2 pts) - “Muscle glycogen can be broken down and sent into the blood to feed the brain.” Correct or incorrect? Explain.
Correct answer	Incorrect. Muscle glycogen is used by that muscle; liver glycogen can help maintain blood glucose for other tissues.
Scoring	1 pt incorrect; 1 pt explains muscle vs. liver glycogen.

Question	6 (2 pts) - Immediate energy needs are met and glycogen stores are full. What happens to excess glucose?
Correct answer	It can be converted to fat through lipogenesis.
Scoring	1 pt fat storage; 1 pt lipogenesis.

Round: The Fast

Question	7 (3 pts) - You have not eaten for several hours. Blood glucose is dropping, but liver glycogen is available. What pathway helps raise blood glucose?
Correct answer	Glycogenolysis: liver glycogen is broken down and glucose is released into blood.
Scoring	1 pt glycogenolysis; 1 pt liver glycogen; 1 pt raises blood glucose.

Question	8 (3 pts) - Fasting continues and glycogen stores are running low. What process makes new glucose from non-carbohydrate sources?
Correct answer	Gluconeogenesis; sources can include glucogenic amino acids, glycerol, and lactate.
Scoring	1 pt gluconeogenesis; 1 pt non-carbohydrate source; 1 pt one correct example.

Question	9 (2 pts) - During prolonged very-low-carbohydrate intake, the liver makes an alternative fuel from fatty acids. Name the process and fuel.
Correct answer	Ketogenesis produces ketones.
Scoring	1 pt ketogenesis; 1 pt ketones.

Round: Glycemic Response

Question	10 (3 pts) - Which is more likely to cause a rapid blood-glucose spike: white bread alone or whole-grain bread with peanut butter? Explain.
Correct answer	White bread alone. Fiber and protein in the second choice slow digestion and glucose absorption.
Scoring	1 pt white bread; 1 pt fiber/protein; 1 pt slower absorption/smaller spike.

Question	11 (2 pts) - Why can whole fruit affect blood glucose differently than fruit juice even though both contain natural sugar?
Correct answer	Whole fruit contains fiber, which slows absorption and produces a more gradual glucose response; juice contains little fiber.
Scoring	1 pt fiber difference; 1 pt slower absorption/lower spike.

Round: Final Boss

Question	12 (5 pts) - Jordan eats pancakes, syrup, and orange juice at 8:00 a.m., then eats nothing for the rest of the day. Describe what happens first and what changes as fasting continues. Include hormones and at least two pathways.
Correct answer	After breakfast, blood glucose and insulin rise. Glucose is used for energy and stored as glycogen; excess may be stored as fat. As fasting continues, insulin falls and glucagon rises. Liver glycogen is broken down by glycogenolysis. As glycogen becomes limited, gluconeogenesis increases; with prolonged low carbohydrate availability, ketogenesis also increases.
Scoring	Up to 5 pts: insulin response; glucose use/storage; glucagon response; glycogenolysis; gluconeogenesis and/or ketogenesis.

Tie-Breaker + Running the Game

Tie-breaker prompt: A student finishes dinner at 7:00 p.m., sleeps overnight, and completes a hard workout at 7:00 a.m. without breakfast. Put these in the most logical order as fasting continues: glycogenolysis, gluconeogenesis, ketogenesis. Then name the hormone most responsible for raising blood glucose early in the fast.

Answer: Glycogenolysis first, followed by increasing gluconeogenesis; ketogenesis becomes more important with prolonged low carbohydrate availability. Glucagon raises blood glucose early in the fast.

Suggested timing: 15-20 minutes total. Use 30-45 seconds for short questions, 60 seconds for 3-point questions, and 2 minutes for the Final Boss.

1. Create groups of 3-5 students and give each group one answer sheet and one pen/pencil.
2. Project the game at the front of the room. Keep this teacher key with you.
3. Show one question at a time. Groups discuss and write one final response before you reveal anything.
4. Click Reveal Answer. Read the scoring line aloud. Groups can self-score, or swap sheets with a neighboring group for scoring.
5. For selected questions, ask one group to explain why before moving on. If many groups miss the same idea, do a 30-second reteach.
6. After Question 12, total the score out of 31. Use the tie-breaker only if needed.

Scoring philosophy: Accept equivalent wording unless the question specifically asks students to name a pathway (glycogenolysis, gluconeogenesis, or ketogenesis).